"Osprey 3.5 V6: Low fuel rail pressure from failing high-pressure pump"

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1 Condition: Low Fuel Rail Pressure on Osprey 3.5 V6

Osprey 3.5 V6 vehicles may exhibit extended cranking, hesitation under load, and a power loss complaint accompanied by an illuminated MIL. The PCM commonly stores DTC P0087 (fuel rail/system pressure too low) and may set P0190 (fuel rail pressure sensor circuit) when measured pressure diverges sharply from the commanded value. Symptoms intensify during hard acceleration when rail pressure demand peaks. For an overview of the affected delivery architecture, see the Osprey Fuel Service Manual :: Fuel Delivery .

2 Cause: Failing High-Pressure Fuel Pump

The dominant cause is mechanical wear of the high-pressure GDI pump, part number HPFP-4120, whose cam-driven plunger loses sealing capacity and cannot sustain commanded rail pressure. In some cases a contaminated or drifting fuel rail pressure sensor FRS-4180 contributes to the P0190 signal-correlation fault. Confirm pump application and supersession against the Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) . Sensor-specific fault behavior is documented in Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults .

3 Diagnosis: Fuel Trim and Pressure Verification

Before parts replacement, verify the complaint by comparing commanded versus actual rail pressure and reviewing long-term fuel trim while monitoring P0087 and P0190 freeze-frame data. Run PROC-FUEL-DIAG following the Osprey Fuel Service Manual :: Fuel Trim and Pressure Diagnosis Procedure to distinguish a failing pump from a faulty FRS-4180 sensor. If the sensor is confirmed out of range, replace it per the Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement . General diagnostic context is in the Osprey Fuel Service Manual :: Fuel System Diagnosis section.

4 Repair Procedure: High-Pressure Fuel Pump Replacement

When the high-pressure pump is confirmed at fault, replace HPFP-4120 following PROC-FUEL-PUMP and observe all fuel-system depressurization and cleanliness precautions. Complete the replacement using the Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure , torquing the high-pressure fittings to specification. After installation, prime the system, clear P0087, and road test to confirm rail pressure tracks the commanded value across the load range.